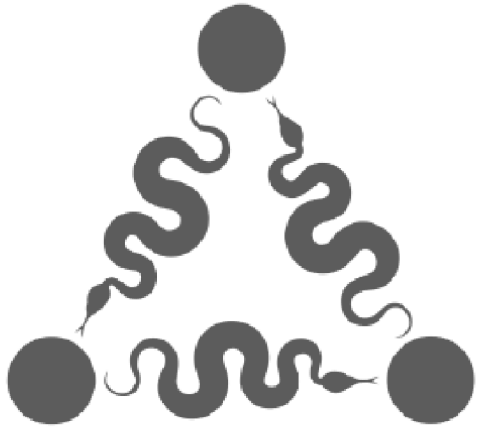
Tutorial on PyKEEN Ecosystem

From Training and Evaluation to Hyperparameter Optimization

Machine Learning

A.A. 2025-2026



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Speaker: Ivan Diliso

Computer Science Department
University of Bari Aldo Moro

Who am I?

Ivan Diliso

Ph.D in Computer Science and Mathematics from **ARA** (Autoamted Learning and Reasoning).

What's my research?

Knowledge Graph Embedding, Neuro-Symbolic AI, Ontologies and Ontology Injection

Prerequisites

Cloud Computing:

- Google Colab You can access all the code materials from this link: GOOGLE COLAB

Local Machine:

- Python 3.x & pip
- Jupyter Notebook
- Visual Studio Code

Contents

Introduction to PyKEEN

What is PyKEEN?

- PyKEEN is a Python library for **training** and **evaluating** knowledge graph embeddings based on **PyTorch**.
- It supports various **models**, **datasets**, and evaluation **metrics**.
- Designed to be user-friendly and extensible.

Key Features:

- Easy-to-use APIs for model training and evaluation.
- Support for a wide range of KG embedding models READTHEDOCS.
- Built-in support for popular datasets READTHEDOCS
- Tools for hyperparameter optimization and result visualization READTHEDOCS

Installation

To get started, you need to install PyKEEN using following command on your **terminal**:

```
pip install pykeen
```

Make sure you have latest Python version installed on your system.

PyKEEN Architecture

1. The **configuration layer** assists users to specify experiments
2. The **learning layer** trains a model with user-defined hyperparameters
3. Finally, **embeddings**, **trained model**, and **configuration settings** are produced

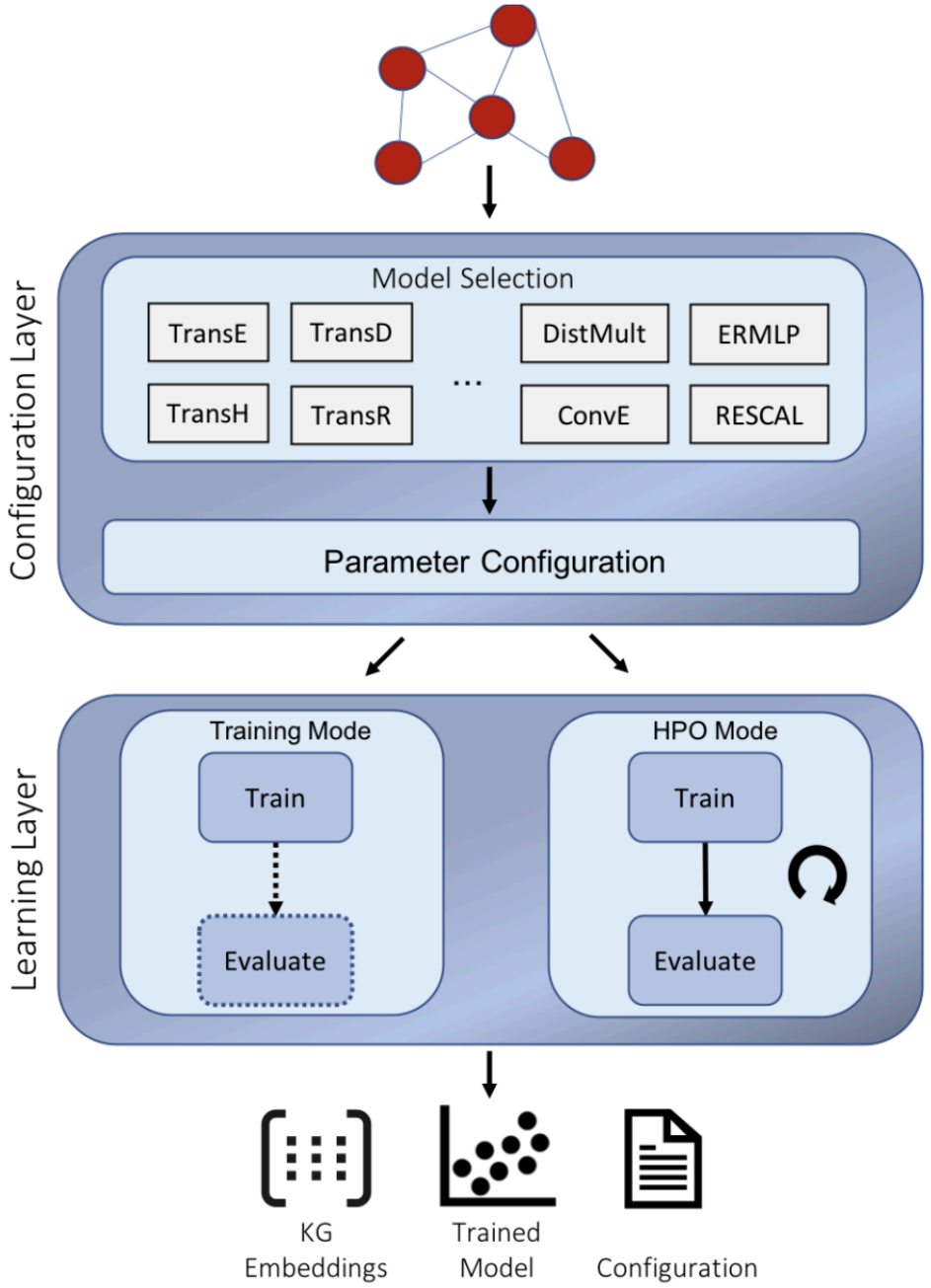


Image from (Ali et. al. 2019)

PyKEEN Architecture

- **CLI**: Run experiments via command line interface
- **Pipeline**: Module starts and controls the configured experiments
- **HPO Pipeline**: Uses **OPTUNA** to optimize the experiments parameters
- **KGE Model**
- **Training**: Training the KGEModel
- **Evaluator**: Evaluate model with defined metrics
- **Inference**: Use the trained model for prediction

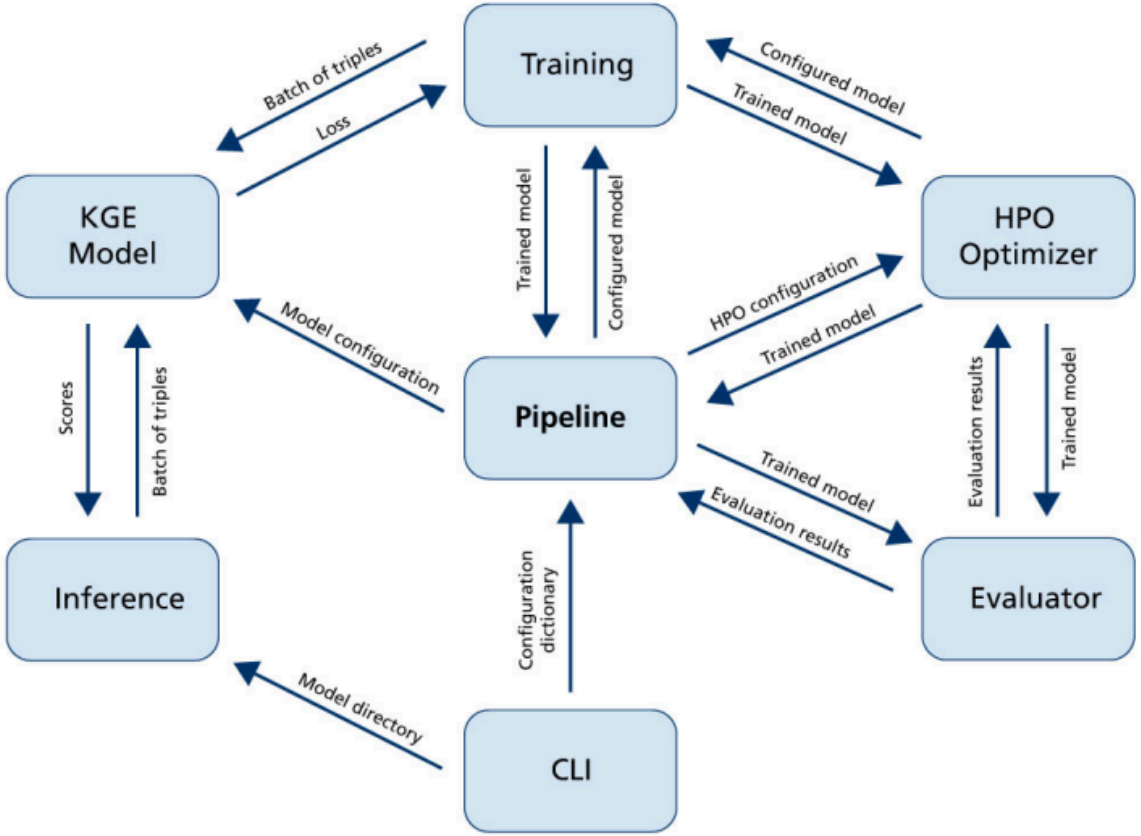


Image from (Ali et. al. 2019)

```
/m/07pd_j       /film/film/genre          /m/02l7c8
/m/06wxw        /location/location/time_zones    /m/02fqwt
/m/012201       /film/music_contributor/film     /m/0ckrnn
/m/05zr0xl      /tv/tv_program/languages         /m/02h40lc
/m/05r6t        /music/genre/artists      /m/04n65n
/m/02jx1        /location/location/contains      /m/01zzk4
```

Creating and Loading a Dataset

PyKEEN supports KGs represented as **RDF** and as **tab-separated values**

```
1       1       5
2       2       6
3       2       3
3       3       4
4       2       6
4       3       6
```

```
Alice   likes   movie
Bob     likes   movie
Charlie likes   movie
David   likes   music
Eve     likes   music
Alice   has     cat
Bob     has     cat
Bob     has     dog
David   has     dog
Eve     has     dog
Frank   has     dog
```

Creating and Loading a Dataset

- PyKEEN supports a variety of datasets for KGE and evaluation tasks. well-known benchmark datasets like **FB15k**, **WN18**, **YAGO3-10**, and more.
- The Library provides tools to easily **load and preprocess these datasets**, facilitating the development and comparison

Lab Activity!
Code on **Colab**

Creating and Loading a Dataset

You can also load a custom dataset directly from file. PyKeen stores data in **TriplesFactory** and automatically assigns an ids to you data in RDF format

```
from pykeen.triples import TriplesFactory
file = Path("path_to_file.txt")
training = TriplesFactory.from_file(file)
```

In this case you need to provide your own **splitting** of data

```
train, valid, test = triples.split(
        ratios = [0.8, 0.1, 0.1],
        random_state = 42
)
```

Training a Model

The **configuration layer** enables users to specify every detail of an experiment for example:

- Datasets
- Execution mode
- KGE model along with its hyper-parameter values
- Negative Sampler
- Details of the evaluation procedure

You can both **instantiate** your own classes or provide **kwargs** for them

Training a Model

There are several possible configurations, PyKeen supports a large pool of Models, Dataset, TrainingLoop, Losses and Metrics

- **Losses**: MarginRankingLoss and BinaryCrossEntropy are the most used
- **TrainingLoop**: sLCWA is the most used for link prediction tasks

Lab Activity!
Code on Colab

Evaluation and Metrics

- The widely applied metrics **mean-rank** and **hits@k** are computed for the evaluation procedure.
- Users can specify whether they want to compute the mean-rank and hits@k in the **raw** or **filtered** setting.
- In the filtered setting, **artificially created negative samples** that are contained as positive examples in the training set will be removed

Lab Activity!
Code on **Colab**

Using Pipelines and Saving Artifacts

The PyKEEN library provides a **high-level abstraction** called the **pipeline**

- Easily to set up and execute the KGE process.
- Essentially a **sequence of steps** that includes:
  - Data loading
  - Model instantiation
  - Training, and Evaluation
- Allows users to perform **end-to-end experiments** without needing to manually write code for each step.

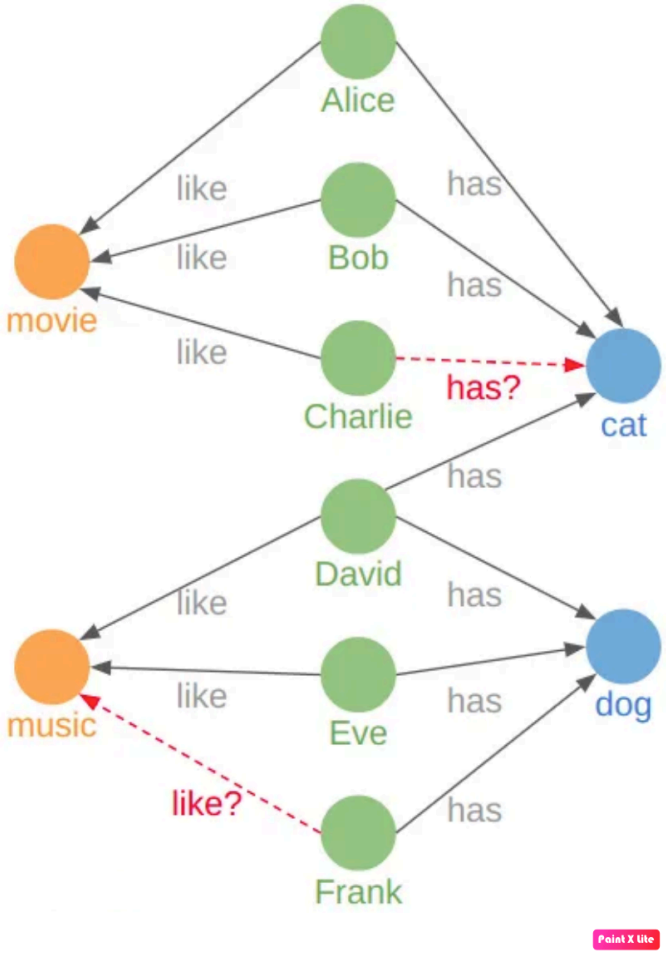
Using Pipelines and Saving Artifacts

After any execution, both from custom training and pipeline, any result can be **saved on disk**, this operation saves both:

- Results
- Model artifacts (weights)
- Running configuration in JSON format
- Metadata
- Training metrics

Lab Activity!
Code on **Colab**

Visualize Learned Embeddings



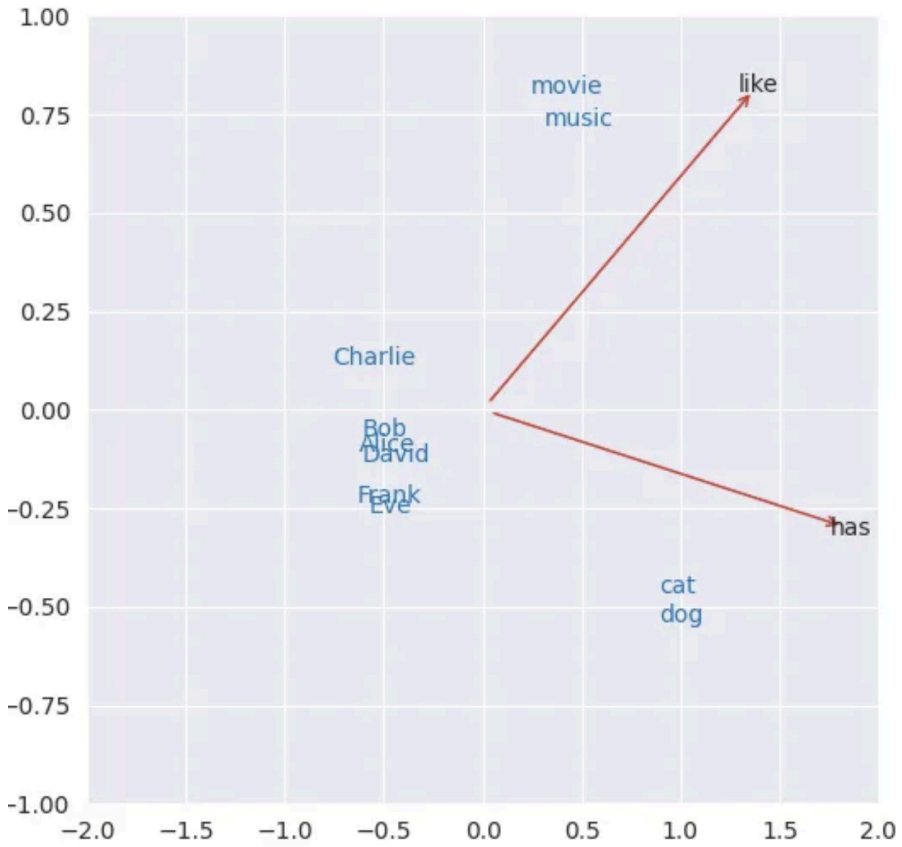
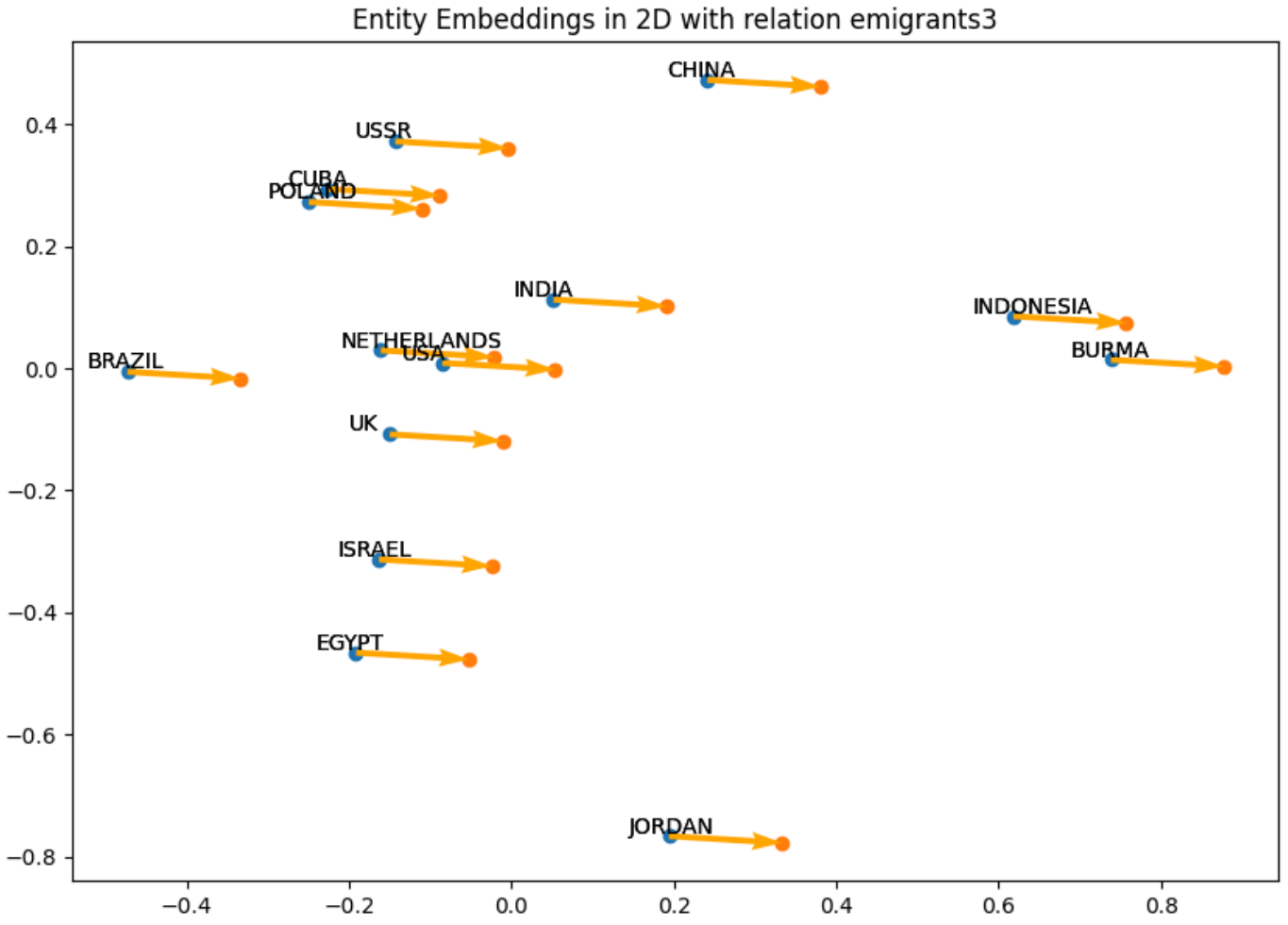


Image Source: https://wasit7.medium.com/tutorial-knowledge-graph-embedding-with-pykeen-22d3b7847cea

Visualize Learned Embeddings

Lets use the learned embeddings to visualize our data!

Lab Activity!
Code on **Colab**

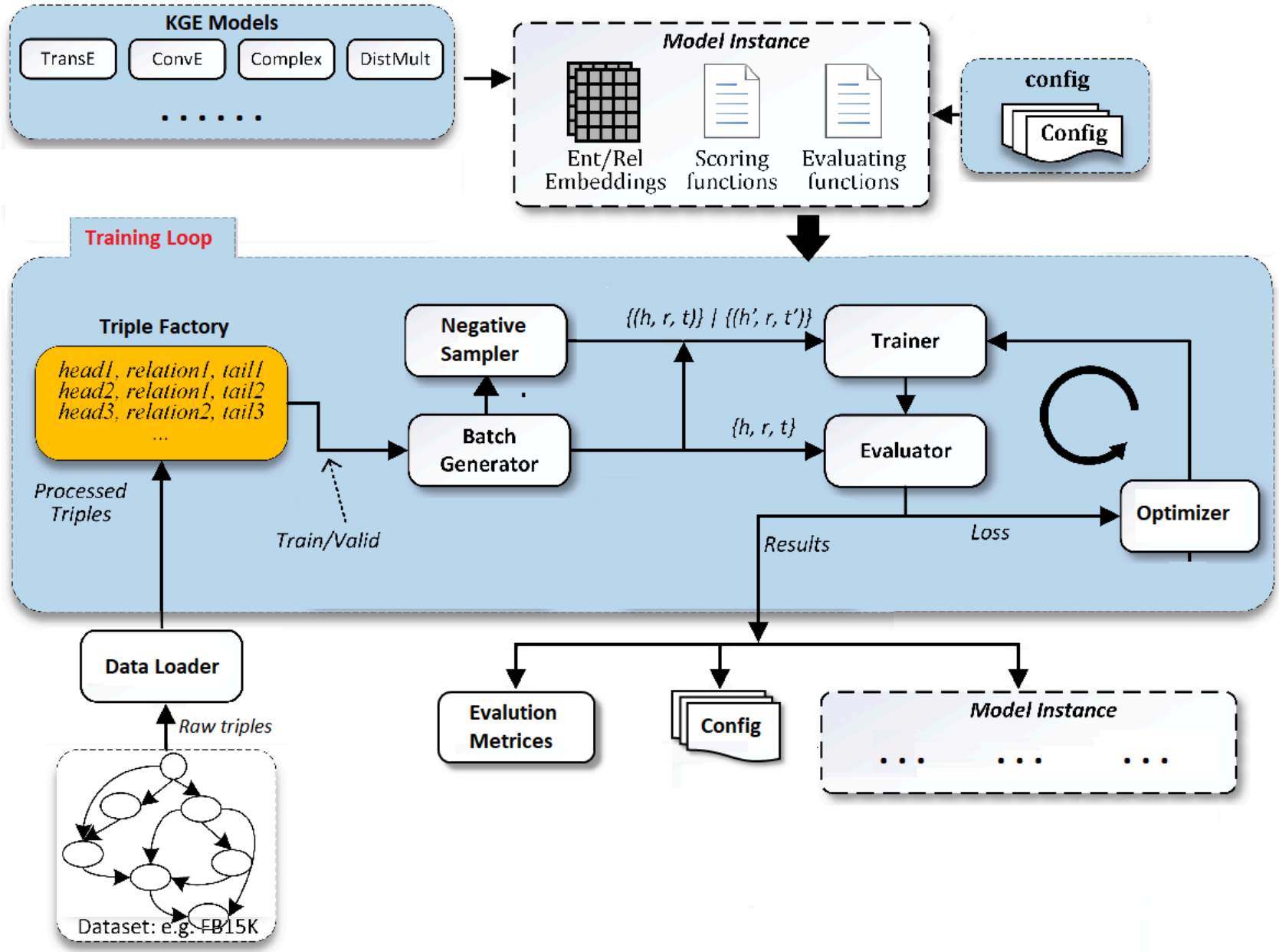


Hyperparameters Optimization

- In training mode users provide for each hyper-parameter the corresponding value.
- In **HPO mode**:
  - You define a set of values for each parameter
  - PyKEEN assists to find **suitable hyper-parameter** values by applying various **searching algorithms** such as random search
  - Uses the validation set to tune and find the optimal parameters

Lab Activity!
Code on **Colab**

Overall Architecture



References

Tips

- Tune hyperparameters for better performance.
- Use visualization tools to understand embeddings.
- Implementing and integrating custom modules for **Research and Development**.

Documentation: READTHEDOCS

Source Code: GITHUB

ARA Group

Thank you for your attention!

Ivan Diliso, Ph.D Student, ARA